CSE276 · ARTIFICIAL INTELLIGENCE FOUNDATIONS

UNIT 4

Evolution of Modern Artificial Intelligence

How the field moved from hand-written symbols to learned statistics - and then to models trained once and reused for everything.

Symbolic vs Statistical AI • Supervised · Unsupervised · Reinforcement • Deep Learning • Foundation Models • Multimodal AI • Agentic AI • Edge and Tiny AI

School of Computer Science & Engineering

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What This Unit Covers - Two Halves

UNIT 4

Part A explains how the field got here. Part B is where it currently is.

- PART A - Evolution of Modern AI
  - Symbolic and statistical Artificial Intelligence
  - Machine learning and its three types: supervised, unsupervised, reinforcement
  - Introduction to deep learning
  - Types and applications of deep learning
- PART B - Foundation Models and Modern Paradigms
  - Foundation models
  - Multimodal Artificial Intelligence
  - Agentic Artificial Intelligence
  - Edge AI and Tiny AI

Symbolic
rules written by people

Statistical
patterns learned from data

Deep
features learned too

Foundation
train once, reuse everywhere

01

SECTION

Symbolic and Statistical AI

IN THIS SECTION

- Two traditions, two answers to the same question
- What each is good at
- Why statistical methods took over
- Why symbolic methods did not disappear

Symbolic AI vs Statistical AI

Not right and wrong - two answers to 'where does the knowledge come from?'

	Symbolic AI	Statistical AI
Knowledge comes from	Human experts, written down as rules	Data, learned automatically
Representation	Symbols, rules, logic, semantic networks	Numbers - weights, probabilities, vectors
Reasoning	Deduction from explicit statements	Inference from patterns in examples
Explainability	High - you can print the chain of rules	Low to moderate - weights are not an explanation
Handles uncertainty	Poorly - logic is mostly binary	Naturally - probability is built in
Handles novelty	Badly - brittle outside its rule set	Better - it generalises from similar examples
Data needed	None - it needs an expert instead	A great deal
Dominant era	1950s to late 1980s	1990s to the present

Why the Field Shifted - and Why Symbolic AI Survives

KEY CONCEPT · **Three things changed, and one requirement did not**

What forced the shift the knowledge acquisition bottleneck. Every rule had to be extracted from a human by hand, and experts frequently cannot articulate what they know.

Then three things arrived together DATA from the internet, COMPUTE from GPUs, and better training ALGORITHMS. Any one alone would not have been enough.

The result learning from data solved perception problems that decades of rule-writing had not.

But symbolic methods survive wherever a decision must be EXPLAINED - medicine, credit, law, safety certification.

And wherever rules are known and exact tax calculation, eligibility, scheduling constraints. Learning those from data would be absurd.

Neuro-symbolic AI the current attempt to combine them: learned perception feeding symbolic reasoning.

DECIDE Calculating income tax owed. Symbolic or statistical?

02

SECTION

Machine Learning and Its Three Types

IN THIS SECTION

- What machine learning is, precisely
- Supervised learning
- Unsupervised learning
- Reinforcement learning

The Three Types of Machine Learning

The paradigm is decided by what feedback is available, not by which algorithm you prefer.

TYPE 1

Supervised learning

- Input: features X WITH labels y for every example
- Goal: learn f such that f(X) ≈ y
- Two families: classification and regression
- Success is directly measurable against held-out labels

TYPE 2

Unsupervised learning

- Input: features X only - no labels at all
- Goal: discover structure already present
- Clustering, dimensionality reduction, anomaly detection
- No ground truth, so evaluation is indirect

TYPE 3

Reinforcement learning

- An agent acts and receives rewards
- Goal: learn a policy maximising long-term reward
- Feedback is delayed, sparse and evaluative
- Nobody says what the right action was

THE DISTINCTION

What feedback you get

- Supervised: 'the answer was cat' - instruction
- Reinforcement: 'that went badly' - evaluation
- Unsupervised: silence
- Three kinds of feedback, three paradigms

Traditional Programming vs Machine Learning

KEY CONCEPT · **The inversion that defines the field**

Traditional programming you supply DATA and RULES; the computer produces ANSWERS.

Machine learning you supply DATA and ANSWERS; the computer produces RULES.

The arrow reverses and that single inversion is the whole idea.

Why it matters some tasks humans perform effortlessly cannot be articulated as rules - recognising a face, understanding speech, telling a cat from a dog.

Polanyi's paradox we know more than we can tell. Machine learning is the response to that gap.

When NOT to use it if a rule works, write the rule. 'Flag every transaction above fifty thousand rupees' needs an IF statement, not a model.

03

SECTION

Deep Learning

IN THIS SECTION

- What makes learning 'deep'
- Why representation learning is the real advance
- The main architectures
- Applications and limitations

What Makes Deep Learning Different

KEY CONCEPT · **The features are learned, not engineered**

The definition neural networks with many hidden layers between input and output.

Classical machine learning a human decides which measurements matter - this is FEATURE ENGINEERING, and it was most of the work.

Deep learning you feed raw input, and the network discovers useful features itself.

The hierarchy in vision, early layers find edges, middle layers find shapes, deep layers find objects. Nobody programmed 'edge' or 'whisker'.

This is the real advance not depth for its own sake, but REPRESENTATION LEARNING - the model builds its own vocabulary.

The cost large data, large compute, and a model whose internal reasoning nobody can read.

Types of Deep Learning and Where They Are Used

Five families, each shaped by the structure of its input.

ARCHITECTURE 1

Convolutional networks (CNN)

- Built for grid data - images above all
- Filters slide across the input, sharing weights
- Medical imaging, quality inspection, face recognition

ARCHITECTURE 2

Recurrent networks (RNN, LSTM)

- Built for sequences - text, speech, time series
- State carried from one step to the next
- Largely superseded by transformers for text

ARCHITECTURE 3

Transformers

- Attention lets every position see every other
- Parallelisable, unlike recurrent networks
- The basis of every modern large language model

ARCHITECTURE 4

Generative models

- GANs, diffusion models, autoencoders
- Produce new data rather than classify existing data
- Image generation, synthetic data, denoising

ARCHITECTURE 5

Graph neural networks

- Built for data with explicit relationships
- Molecules, social networks, road networks
- Recommendation and drug discovery

THE LIMITATION

What they all share

- Large data and compute requirements
- Poor interpretability
- Confident failure outside the training distribution

CHECK YOUR UNDERSTANDING

Quick Check - Parts A

UNIT 4

QUICK CHECK

A hospital must be able to explain, rule by rule, why a treatment was recommended. Which approach fits best?

A. A deep neural network, because accuracy matters most

B. A symbolic rule-based system, because its reasoning chain is inspectable

C. Unsupervised clustering of patient records

D. Reinforcement learning from patient outcomes

ANSWER: B The binding requirement is EXPLAINABILITY, not raw accuracy. A production-rule system can print the chain of rules that led to its conclusion, which a clinician can inspect and challenge. This is why symbolic methods persist in regulated domains despite being older technology.

04

SECTION

Foundation Models

IN THIS SECTION

- What a foundation model is
- Pre-training and adaptation
- Why the economics changed
- Risks of building everything on one model

Foundation Models - Train Once, Adapt Many Times

KEY CONCEPT · A change in how AI systems are built, not just in accuracy

The definition a very large model trained on a broad corpus, then ADAPTED to many downstream tasks rather than trained from scratch for each.

How it is trained self-supervised - the model creates its own labels from the data, typically by predicting hidden parts of the input. No annotation is needed.

Why that matters it removes the labelling bottleneck, so training can use essentially unlimited raw text or images.

Adaptation fine-tuning on a small labelled set, or simply prompting with instructions and examples.

The economic change the expensive training happens once, centrally. Everyone else adapts - which puts capable models within reach of small teams.

Examples large language models, vision-language models, speech models, and protein-structure models.

The risks of the same foundation everywhere

- Homogenisation - one model's biases propagate into every application built on it
- A single point of failure across an entire ecosystem
- Concentration - very few organisations can afford to train one

Four Modern AI Paradigms

Two paradigms about capability, and two about where the computation physically happens.

PARADIGM 1

Multimodal AI

- One model handling text, images, audio and video together
- Closer to how perception actually works
- Describe an image, answer questions about a chart
- Blurs the neat single-modality categories

PARADIGM 2

Agentic AI

- Models that PLAN and ACT over multiple steps
- Call tools, use results, decide the next step
- Unit III's agent ideas, built on a foundation model
- Open problem: reliability over long action chains

PARADIGM 3

Edge AI

- Inference on the device, not in the cloud
- Lower latency, works offline, data stays local
- Phones, cameras, vehicles, industrial sensors
- Constrained by memory and power

PARADIGM 4

Tiny AI (TinyML)

- Models small enough for microcontrollers
- Kilobytes of memory, milliwatts of power
- Keyword spotting, vibration monitoring, wearables
- Quantisation and pruning make it possible

Edge AI and Tiny AI - Why Move the Computation?

The same trade-off in both: accept a smaller model to gain independence from the network.

Reasons to run on the device

Latency no network round trip. A vehicle cannot wait for a server to answer.

Availability it works with no connectivity - a farm, a mine, a moving train.

Privacy raw data never leaves the device, which matters for health, home and camera data.

Cost no per-request inference bill and no bandwidth charges at scale.

Energy sending data over a radio often costs more power than computing locally.

What it costs

Model size must fit in device memory - sometimes only kilobytes.

Accuracy quantisation and pruning shrink the model and usually cost some accuracy.

Updating pushing a new model to a million devices is harder than updating one server.

Monitoring you cannot easily observe what a deployed edge model is doing.

The trade you accept a smaller, slightly worse model in exchange for latency, privacy and independence.

The Evolution in One Line

Each era answered the previous era's bottleneck.

1956-80s	1990s	2012	2017-20	2022-now
Symbolic AI	**Statistical AI**	**Deep learning**	**Transformers**	**Foundation & agentic**
Rules written by experts	Learning from data	Features learned, not engineered	Attention replaces recurrence	Train once, adapt everywhere
Search and logic	Probability and ML	AlexNet and GPUs	Scales with data and compute	Multimodal and agentic
Expert systems	Feature engineering by hand	Perception largely solved	Foundation models become possible	Pushing outward to the edge

UNIT IV CONSOLIDATION

Activity: Place the System

UNIT 4

GROUP WORK · 12 minutes

In groups of 3-4, take four AI systems and place each on the evolution timeline - then justify the placement.

1

The four systems

- A spam filter using hand-written keyword rules
- A hospital triage model trained on 50,000 labelled records
- A phone that unlocks with your face, entirely offline
- An assistant that reads your email and books a meeting

2

For each, decide

- Symbolic, statistical, deep, foundation or agentic?
- Which type of machine learning, if any?
- Cloud or edge, and why?

3

Then argue

- Which one could have been built in 1990, and which could not?
- What specifically was missing before?

4

The awkward question

- Which of the four would you NOT build with a foundation model, and why?

DELIVERABLE One speaker per group: 60 seconds on the system your group found hardest to place, and why it resisted classification.

UNIT IV RECAP

Unit IV - The Seven Ideas Worth Remembering

UNIT 4

1. **Symbolic and statistical AI answer the same question differently**
   Symbolic gets its knowledge from experts as rules; statistical learns it from data. Neither is obsolete.

2. **The knowledge acquisition bottleneck forced the shift**
   Experts cannot always articulate what they know, and hand-writing rules does not scale.

3. **Machine learning inverts traditional programming**
   Data plus rules gives answers; data plus answers gives rules. That inversion is the whole idea.

4. **The three types differ by the feedback available**
   Supervised gets instruction, reinforcement gets evaluation, unsupervised gets silence.

5. **Deep learning's advance is representation learning**
   The features are discovered by the model rather than engineered by a human. Depth is the means, not the point.

6. **Foundation models change the economics**
   Train once on broad data with self-supervision, then adapt cheaply to many tasks - at the cost of homogenised risk.

7. **Edge and Tiny AI move computation to the device**
   Trading model size and some accuracy for latency, privacy, offline operation and cost.

NEXT UP Unit V - Generative AI and Responsible AI: large language models, prompt engineering, RAG and hallucination; then explainability, fairness, privacy and governance.